

Dr. Leyna May-Gurr

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SUMMARY

Software engineer with a Mathematics PhD, currently building full-stack features at a women's fitness startup. Strong in Python and JavaScript through projects spanning real-time multiplayer and 3D physics simulation, plus a published recursive graph algorithm. A confident communicator through running seminar series, organising international conferences, and Women in Tech workshops.

Based in Munich and eligible to work in Germany without visa sponsorship.

EDUCATION

University of Sheffield 2025

PhD in Mathematics. Thesis 'Hidden Symmetries, Trees and Operads'

Durham University 2020

Masters in Mathematics, First class honours. Thesis 'Morse Theory: Morse-Bott Functions and Homology'

EXPERIENCE

Software Engineer 2026 – Present

Bobbi, women's fitness tracking app

- Second developer on a two-person engineering team
- Building the workout guide pages in JavaScript and Vue.js

Graduate teacher 2021 – 2025

University of Sheffield

- Taught Python and R to undergraduate engineering students; led tutorials in mathematics and engineering
- Marked examinations and assessments, reviewing and debugging student code with constructive feedback
- Supervised a UKRI-funded undergraduate summer project on knot theory (June–July 2023)

PROJECTS

Ultimate Noughts and Crosses | *Python, FastAPI, JavaScript, Vue.js, HTML, CSS* 2026

- Built a real-time multiplayer “ultimate” variant, nine boards nested inside one, with a Vue.js front-end and a FastAPI backend handling WebSocket connections, nested game state, and room creation/joining
- Deployed to Oracle Cloud via GitHub Actions; playable at noughts-and-crosses.leynamay.com, source at github.com/LeynaM/noughts-and-crosses

Tilt Game | *JavaScript, Three.js, React, Node.js, SQLite, HTML, CSS* January 2025

- Developed a full-stack physics-based browser game using Three.js, implementing custom physics algorithms for ball movement, platform tilting, and collision detection
- Built mouse controls, stored scores in SQLite behind a React leaderboard, and deployed to Oracle Cloud via GitHub Actions; playable at tilt.leynamay.com, source at github.com/LeynaM/tilt-game

PUBLICATIONS

Quartic graphs which are Bakry-Émery curvature sharp | *Python* March 2020

- Designed and implemented a recursive algorithm to generate unique graphs with certain curvature properties; published in Discrete Mathematics, Volume 343, Issue 3 (doi.org/10.1016/j.disc.2019.111767)

SKILLS

Frontend: JavaScript, Vue.js, React, HTML, CSS

Backend: Python, FastAPI, Node.js, SQLite

Tools: Git, Linux, LaTeX | **Languages:** English (Native), German (B1)